

Chapter 1

Foundation of Engineering Economy

Engineering Economy: **decide** where and how to invest **capital funds** to **add value** in the **future** by:

- Formulating, estimating, and evaluating **expected** economic outcomes of **alternatives**.

Engineering economy is a collection of mathematical techniques that simplify the economic evaluation of alternatives. This is valid not only for business and engineering projects, but also **personal finances**.

Your Role: “Engineers play a vital role in capital investment **decisions** based upon their ability and experience to **design, analyze, and synthesize**”.

Examples of decisions an Engineer can answer:

- Should we upgrade equipment to improve productivity?
- Or should we replace it?
- Or should we do nothing?
- Should we upgrade from supplier A or B?
- Buy a new phone, car, house, etc

1.1 Steps for a study

Real-world engineering problems are rarely neatly packaged with all the necessary numbers provided. Before jumping into compound interest formulas and cash flow diagrams, an engineer must understand the broader context of decision-making. Tackling a capital investment problem requires a systematic, structured approach to ensure no critical factors are overlooked.

The standard procedure for an engineering economy study consists of the following nine steps:

1. Identify and understand the problem; define the objective¹.
2. Collect relevant data and information.
3. Develop alternatives and estimate costs
4. Define the evaluation criteria ²
5. Evaluate each alternative
6. Select the best alternative
7. Implement the solution
8. Monitor the results
9. refine the solution (go back to 3)

To better understand this workflow, we can group these nine steps into three distinct phases:

Phase 1: The Foundation (Steps 1–3) This is often the most difficult and time-consuming part of a project. A perfectly executed mathematical analysis is useless if it solves the wrong problem.

- **Gathering Data:** In textbooks, data is perfectly packaged for you. In the real world, Step 2 requires digging through historical maintenance logs, calling vendors for equipment quotes, and projecting future inflation rates.

¹“A problem well stated is a problem half solved” - Charles Kettering

²We focus in this course on economic criteria (e.g.: present worth, future worth, rate of return, payback period, etc.) However, non-economic factors (such as resource availability, environmental impact) are also important

- **Alternatives:** Step 3 requires creativity. Remember that the “Do Nothing” (status quo) option is almost always a valid alternative that must be evaluated against new investments.

Phase 2: Mathematical Modelling and Evaluation (Steps 4–6) This phase involves translating physical alternatives (like building a new facility versus upgrading an old one) into a timeline of expected cash inflows and outflows. Once the costs and cash flows are mapped, we apply our chosen economic criteria (Step 4) to evaluate them fairly (Step 5) and objectively select the best path forward (Step 6).

Phase 3: Action and Accountability (Steps 7–9) Engineering does not end when a decision is made. Once management selects and implements the best alternative, engineers must track its real-world performance. For example, if a new machine was estimated to save \$2,000 a month in electricity, Step 8 involves checking the actual utility bills a year later. Step 9 uses that feedback to either refine the current operation or improve the accuracy of future estimates.

While engineering economy principles are applied most extensively in Steps 2 through 6, a successful professional engineer must be prepared to engage with the entire lifecycle of a project.

1.2 Time Value of Money

“It is a well-known fact that **money makes money**. The **time value of money** explains the change in the amount of money over time for funds that are owned (invested) or owed (borrowed). This is the most important concept in engineering economy.”

If we invest money, we expect to have the original amount plus some interest. If we borrow money, we expect to return the original amount plus some interest.

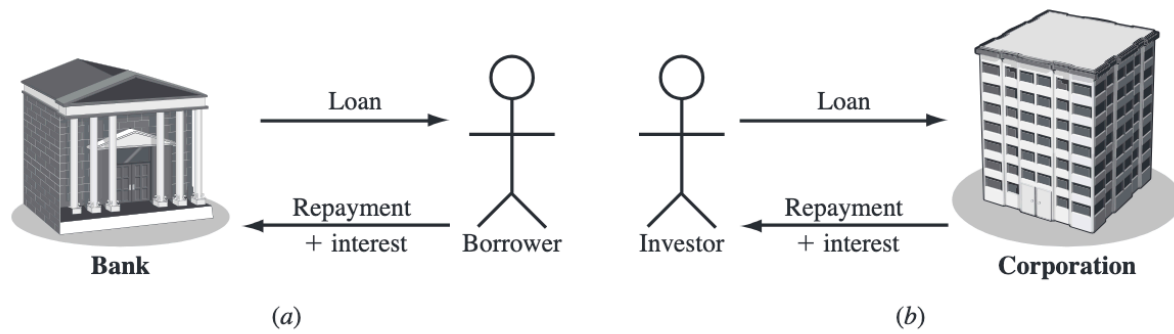
Money possesses a **time value**

Interest and inflation are the most significant drivers of the Time Value of Money.

1.3 Interest

- Interest is paid when borrowing money.

$$\text{interest} = \text{final loan amount} - \text{principal}$$

**Figure 1-2**

(a) Interest paid over time to lender. (b) Interest earned over time by investor.

Figure 1.1: Interest from two point of view

The **interest rate** is:

$$\begin{aligned} \text{interest rate (\%)} &= \frac{\text{interest accrued per time unit}}{\text{principal}} \times 100\% \\ &= \frac{\text{final loan amount} - \text{principal}}{\text{principal}} \times 100\% \end{aligned}$$

- Interest is earned when saving, investing, or lending money.

$$\text{interest} = \text{total investment value} - \text{principal}$$

The **rate of return** is:

$$\begin{aligned} \text{Rate of return (\%)} &= \frac{\text{interest accrued per time unit}}{\text{principal}} \times 100\% \\ &= \frac{\text{total investment value} - \text{principal}}{\text{principal}} \times 100\% \end{aligned}$$

The **time unit** for both interest rate and rate of return is called the **interest period**.

Example 1.1. Finn borrows \$100,000 and must repay a total of \$108,000 exactly one year later. Determine the interest amount and the interest rate paid.

- interest paid = \$108,000 - \$100,000 = \$8,000
- interest rate = $\frac{\$8,000}{\$100,000} = 8\%$ per year

Example 1.2. Bora plans to borrow \$3,000 from a bank for 1 year at 12% interest for

new computer. Compute the interest and the total amount due after 1 year.

- interest = $\$3,000 \times 0.12 = \360
- Total loan value = $\$3,000 + \$360 = \$3,360$

Example 1.3. Karel wants to deposit an amount such as after a year it would have a value of \$5,000 at an interest rate of 6% a year. 1) What is the principal required to achieve this future value. 2) Calculate the amount of interest earned during this year period.

1. The total amount accrued after one year is the sum of the principal, denoted by x , and the earned interest.

$$\begin{aligned}\text{total accrued} &= \text{principal} + \text{principal} \times \text{interest rate} \\ \$5000 &= x + x \times 0.06 = 1.06x\end{aligned}$$

The principal (original deposit) is:

$$x = \frac{5000}{1.06} = 4716.98$$

2. Interest earned = $5000 - 4716.98 = 283.02$

1.4 Economic Equivalence

Definition – Economic Equivalence

Different sums of money at different times are equal in value

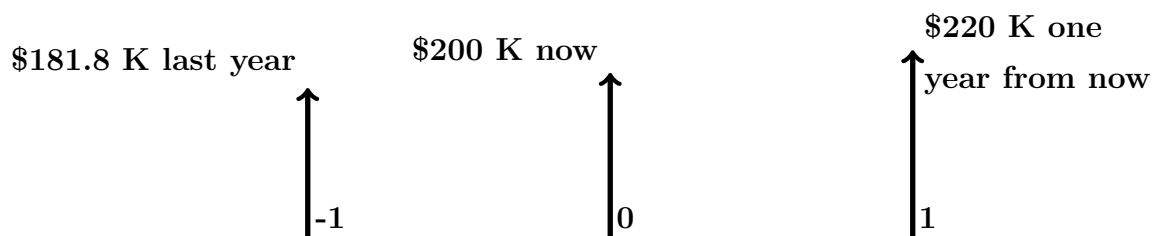


Figure 1.2: The three sums of money that are equivalent **given** an interest rate of 10% per year

Example 1.4. You want to replace your iPhone. A new one this year is \$1100 and it is estimated that the next year's model will be \$1200. At an interest rate of (1) 6% (2) 10%, would you replace it this year or the next one?

1. \$1,200 next year is equivalent to $1200/1.06 = \$1,132$.

We can say that the paying 1200 next year is **equivalent** to paying 1132 this year, which is still \$32 more than buying this year. Hence, it is more economical to buy this year and save \$32.

This is a **Present Worth** analysis.

OR

1100 this year is equivalent to $1100 \times 1.06 = 1166 < 1200$.

We can say that buying an iPhone this year would save us \$34 as compared to buying next year at 1200.

This is a **Future Worth** analysis

Why is there a difference of \$2 between present and future worth analysis?

This difference stems again from point of time we are looking at the problem. When we analyzed the Present worth of alternatives, the savings were around \$32. The Future worth of this “\$32” is $32 \times 1.06 \approx 34$.

2. \$1,200 next year is equivalent to $1200/1.1 = \$1,090.9$. It is better to wait till next year and save 9 dollars.

Note – Present Worth and Future Worth

Present Worth and Future Worth are fundamental concepts that we will define and explore extensively in upcoming chapters. For now, simply recognize them as ways to shift money backward or forward in time.

1.5 Simple and Compound Interest

So far we have analyzed problem over a single interest period. When we have multiple periods, we need to differentiate two definitions of interest.

Definition – Simple vs. Compound Interest

Simple interest is calculated using the **principal only**, while **compound interest** is calculated on the **principal plus the total amount of interest accumulated in all previous period**.

1.5.1 Simple interest

With simple interest, the borrower pays interest only on the borrowed amount (the principal). Let P be the borrowed amount, n be the number of periods, and i be the interest rate per period. The total amount F the borrower pays back n periods later is:

$$\begin{aligned} F &= P + \underbrace{iP + \cdots + iP}_{n \text{ times}} \\ &= P + niP \\ &= P(1 + ni) \end{aligned}$$

In practice, **simple interest is rarely used** because it ignores the **time value of money**. Lenders expect a return on all capital tied up in a loan—including unpaid interest—making compound interest the commercial standard.

1.5.2 Compound Interest

With compound interest, the borrower pays on the borrowed amount plus interest accumulated in previous periods. In other words, **the borrower pays interest on the interest**.

Let F_t be the future value at the end of period t . The future value at the end of the first period is:

$$F_1 = P + Pi = P(1 + i)$$

End of Period 2:

$$F_2 = F_1 + F_1i = F_1(1 + i) = P(1 + i)(1 + i) = P(1 + i)^2$$

End of period 3:

$$F_3 = F_2(1 + i) = P(1 + i)^2(1 + i) = P(1 + i)^3$$

End of Period n :

$$F_n = F_{n-1}(1 + i) = P(1 + i)^{n-1}(1 + i) = P(1 + i)^n$$

Example 1.5. A manufacturing facility plans to upgrade its production line by purchasing an automated packaging machine. The plant manager needs to secure a loan of

\$100,000 for a duration of 5 years and is evaluating two different financing offers.

- A private partner offers the full amount at an annual simple interest rate of 8.5%.
- A specialized industrial financing firm offers the loan at an annual compound interest rate of 6.0%.

Option A: Private Investor (Simple Interest)

$$\begin{aligned}
 P &= \$100,000 \quad i_s = 0.085 \text{ per year} \quad n = 5 \text{ years} \\
 F_{\text{simple}} &= P(1 + ni_s) \\
 &= 100,000(1 + 5 \times 0.085) \\
 &= 100,000(1.425) = \$142,500
 \end{aligned}$$

The total interest paid to the private investor is \$42,500.

Option B: Specialized Industrial Bank (Compound Interest)

$$\begin{aligned}
 i_c &= 0.06 \text{ per year} \\
 F_{\text{compound}} &= P(1 + i_c)^n \\
 &= 100,000(1 + 0.06)^5 \\
 &= 100,000(1.338226) \approx \$133,822.56
 \end{aligned}$$

The total interest paid to the bank is \$33,822.56.

Conclusion:

Even though the bank's loan involves compound interest (meaning the borrower pays interest on accumulated interest), the lower annual rate (6% vs. 8.5%) makes it the more economical choice over a 5-year period. The plant saves \$8,677.44 by choosing the compound interest offer from the specialized firm.

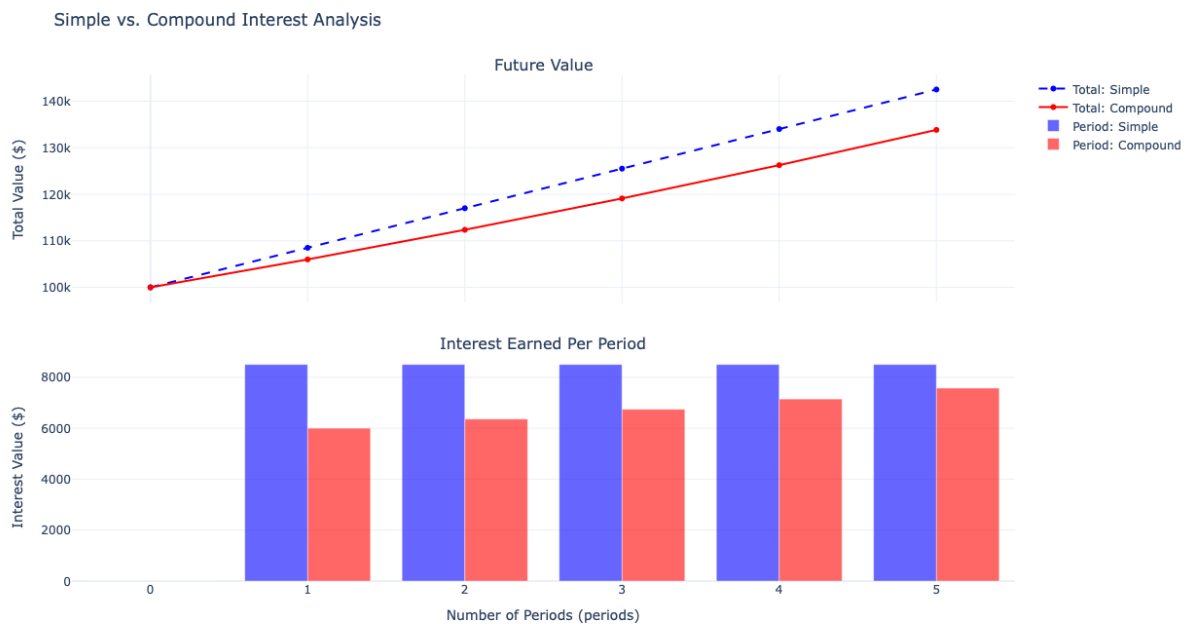


Figure 1.3: Evolution of Future Value of the loan (upper), and the interest accumulation (lower). See online on Colab

1.6 Cash Flows

Definition – Cash Flows

Cash flows are the amounts of money estimated for future projects or observed for project events that have taken place

1.6.1 Cash Inflow

Definition – Cash Inflows

Cash inflows are the receipts, revenues, incomes, and savings generated by project and business activity.

Amount of funds flowing into a firm.

A plus sign indicates a cash inflow.

Example of cash inflows:

- Sales revenue: +\$100,000 from selling equipment
- Borrowed money: +\$50,000 to install solar power system
- Savings: +\$2,000 per month electricity cost
- Asset salvage value: +\$15,000 from selling the CNC machine at the end of its 5-year

useful life

- Income tax savings: +\$4,500 annually due to equipment depreciation deductions

1.6.2 Cash Outflows

Definition – Cash Outflows

Cash outflows are costs, disbursements, expenses, and taxes caused by projects and business activity.

Amount of funds flowing out of a firm.

A negative sign indicates a cash outflow.

Example of cash outflows:

- Equipment purchase: -\$50,000 to acquire an energy management system
- Installation costs: -\$5,000 for site preparation and system integration
- Operating and maintenance: -\$1,500 per month for tooling, lubrication, and routine servicing
- Labor expenses: -\$4,000 - \$8000 per month for hiring a specialized machine operator
- Loan repayment: -\$22,000 annually for principal and interest payments to the industrial bank
- Income tax liability: -\$8,000 annually due to increased taxable profits from higher production capacity

Remark – Point vs. Range Estimates

We can note that future costs could be given in the form of a **point estimate** or a **range estimate**. The main criteria for which kind of estimate to give depends on how **confident** the estimates are. Higher confidence in the estimates make the use of a point estimate valid. However, as soon as an estimate cannot be provided accurately, point estimates are not preferred. For instance, this could be caused by the lack of knowledge and expertise of the person doing the estimation, or it could be because of the uncertainties in external factors, such as shipping.

Once all cash inflows and outflows are estimated (or determined for a completed project), the net cash flow for each time period is calculated:

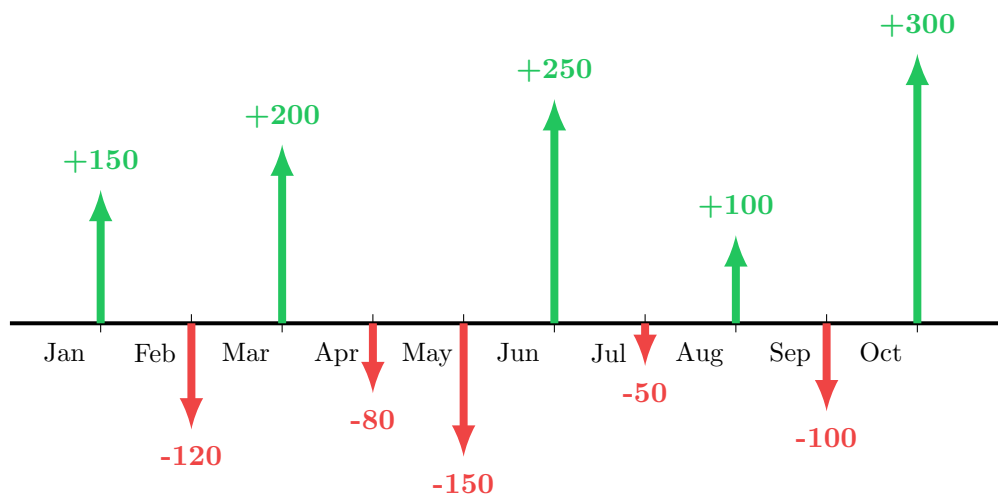
$$\text{Net flow} = \text{cash inflows} - \text{cash outflows}$$

1.6.3 Cash flow diagram

The cash flow diagram is a very important tool in an economic analysis, especially when the cash flow series is complex. As shown in Figure 1.4, the cash flow diagram is a **graphical representation** of cash flows drawn on the y-axis with a timescale on the x-axis. The diagram includes what is known, what is estimated, and what is needed. That is, once the cash flow diagram is complete, another person should be able to work the problem by looking at the diagram.

A cash flow diagram has the following characteristics:

- The horizontal line is a time scale, with progression of time moving from left to right.
- The arrows signify cash flows and are placed at the **end of the period**. Downward arrows represent expenses (negative cash flows or cash outflows) and upward arrows represent receipts (positive cash flows or cash inflows).
- The cash-flow diagram is dependent on the point of view.



Cash flows are shown as directed arrows
(+) for inflow (-) for outflow)

Figure 1.4: Sample of cash flow diagram with monthly period

Example 1.6. In a company's renovation of a small office building, two feasible alternatives for upgrading the heating, ventilation, and air conditioning (HVAC) system have been identified. Either Alternative A, that is to rebuild (overhaul) the existing HVAC system or Alternative B, that is to install a new HVAC system that utilizes existing ductwork must be implemented. The costs are shown in Table 1.1: At the end of eight

	Alternative A	Alternative B
Equipment, labor, and materials	18,000	60,000
Annual cost of electricity	32,000	9,000
Annual maintenance expenses	2,400	16,000
Replacement of a major component four years	-	9,400

Table 1.1: Cost elements of the two alternatives in USD (\$)

years, the estimated market value for Alternative A is 2,000 and for Alternative B it is 8,000. **Assume** that both alternatives will provide comparable service (comfort) over an eight-year period, and **assume** that the major component replaced in Alternative B will have no market value at EOY eight.

1. Use a cash-flow table and end-of-year convention to tabulate the net cash flows for both alternatives.
2. Determine the annual net cash-flow difference between the alternatives (B - A).

	A	B	C	D	E
1		Alternative A	Alternative B	Difference	Cumulative
2	End of Year	Net Cash Flow	Net Cash Flow	(B-A)	Difference
3	0 (now)	-18000	-60000	-42000	-42000
4	1	-34400	-25000	9400	-32600
5	2	-34400	-25000	9400	-23200
6	3	-34400	-25000	9400	-13800
7	4	-34400	-34400	0	-13800
8	5	-34400	-25000	9400	-4400
9	6	-34400	-25000	9400	5000
10	7	-34400	-25000	9400	14400
11	8	-32400	-17000	15400	29800
12	Total	-291200	-261400		29800

Figure 1.5: Table showing the cash flow following end of year convention for both alternatives

From the viewpoint of the company, it has two alternatives and one of them must be selected. Hence, choosing nothing is not an option. The results from the spreadsheet shown in Figure 1.5 highlight the following:

- Alternative B is the recommended choice because it is \$29,800 more **cost-efficient**.
- It takes 6 years for the extra \$42,000 investment in Alternative B to generate sufficient savings in annual expenses to justify the higher investment.

However, in this example we **excluded the time value of money**. What would be our conclusion when we include that?³

1.7 Minimum Attractive Rate of Return

For any investment to be profitable, the investor (corporate or individual) expects to receive more money than the amount of capital invested. In other words, a **fair** rate of return (ROR), or return on investment, must be realizable.

Remember that the definition of ROR is the amount earned divided by the principal:

$$\begin{aligned}\text{Rate of return (\%)} &= \frac{\text{interest accrued per time unit}}{\text{principal}} \times 100\% \\ &= \frac{\text{total investment value} - \text{principal}}{\text{principal}} \times 100\%\end{aligned}$$

Alternatives are evaluated upon the projection that a **reasonable ROR** can be **expected**⁴.

Definition – Minimum Attractive Rate of Return (MARR)

The Minimum Attractive Rate of Return (MARR) is a reasonable rate of return established for the evaluation and selection of alternatives. A project is not economically viable unless it is expected to return at least the MARR. MARR is also referred to as the hurdle rate, cutoff rate, benchmark rate, and minimum acceptable rate of return.

The MARR is usually a policy issue resolved by the top management of an organization in view of numerous considerations. Among these considerations are the following:

1. **Cost and Source of Capital:** The amount of money available for investment, and whether those funds are equity⁵ or debt⁶.
2. **Project Portfolio:** The number of good projects available and their urgency (i.e., whether they sustain essential present operations or are elective expansions).
3. **Project Risk:** The amount of perceived risk associated with the investment, along with the administrative costs over short versus long planning horizons. Riskier

³To be discussed in Chapter 2

⁴We expect something to happen because we are uncertain of the outcome of the investment

⁵A company uses its own assets to finance, often through issuing stocks

⁶A company borrows money to finance, often through issuing bonds

projects require a higher MARR.

4. **Organization Type:** Whether the entity is a government, public utility, or private industry (governments typically have much lower MARRs than private startups).

Why isn't the MARR just 0%?

You might wonder: If a project makes even 1% profit, isn't that better than nothing? Why set the hurdle so high?

Companies set the MARR higher than zero because **money isn't free**. For example, if a company has \$100,000, they could simply put it in a guaranteed savings account earning 5% interest, yielding \$5,000 per year risk-free. Therefore, any project they invest that \$100,000 into *must* return significantly more than 5% to justify the risk and effort. If a proposed project only returns 1%, it is a net loss compared to just leaving the money in the bank.

If you want to purchase a new widescreen HDTV, but do not have sufficient money (capital), you could choose one of the following alternatives:

- Obtain a bank loan for, say, a cost of capital of 9% per year and pay for the TV in cash now.
- Use your credit card and pay off the balance on a monthly basis. This approach will probably cost you at least 15% per year.
- Use funds from your savings account that earns 5% per year and pay cash. This approach means that you also forgo future returns from these funds⁷.

The 9%, 15%, and 5% rates are your **cost of capital** estimates to raise the capital for the system by different methods of capital financing.

Definition – Sources of Capital

In general, capital is developed in two ways:

Equity financing: The firm uses its own money. This includes cash on hand or selling partial ownership of the company (issuing stocks).

Debt financing: The firm uses borrowed money. This includes taking loans from a bank or borrowing from investors with a promise to pay them back with interest (issuing bonds).

⁷In engineering economy, this lost potential return is called an **opportunity cost**

The MARR is set such that:

$$\text{MARR} > \text{cost of capital}$$

Combining these two ways is widely done in practice. For instance, you can buy the TV with 40% credit card money at 15% per year and 60% from the savings funds that is earning you 5% per year. In that particular example, we calculate the **weighted average cost of capital** (WACC) as follows:

$$\begin{aligned}\text{WACC} &= (0.40 \times 15\%) + (0.60 \times 5\%) \\ &= 6\% + 3\% \\ &= 9\% \text{ per year}\end{aligned}$$

Therefore, your blended cost of capital is 9%, and your MARR must be set higher than this threshold.

A project is viable if and only if its ROR $\geq$ MARR.

1.8 The Rule of 72: A Mental Math Shortcut

While modern engineering economy relies on precise compound interest formulas and software, professionals often need a way to evaluate investments rapidly during meetings or site visits. For this, we use the **Rule of 72**.

Definition – Rule of 72

The **Rule of 72** is a mathematical approximation used to estimate how long it will take for an investment (or a debt) to exactly double in value at a given fixed annual compound interest rate.

Instead of solving the logarithmic compound interest equation, you simply divide the number 72 by the annual interest rate (expressed as a whole number, not a decimal):

$$\text{Years to Double} \approx \frac{72}{\text{Interest Rate}}$$

Example: If an industrial project guarantees a 9% annual rate of return, how long will it take for the invested capital to double?

$$\text{Years to Double} \approx \frac{72}{9} = 8 \text{ years}$$

(Note: The exact mathematical answer is 8.04 years. The Rule of 72 is remarkably accurate for typical interest rates between 5% and 12%.)

When to Use the Rule of 72

The Rule of 72 is an estimation tool, meaning it should **never** be used for final engineering budgets or formal project approvals.

However, it is highly valuable in the following scenarios:

- **Initial Project Screening:** If a manager states that a \$1 million automation upgrade must double its value in 5 years, the engineer instantly knows the project needs to yield roughly $72/5 = 14.4\%$ annually. If the best projection is only 8%, the project is immediately rejected without wasting time on detailed spreadsheets.
- **Understanding the "Weight" of the MARR:** Stating a corporate MARR is "12%" can feel abstract. Using the Rule of 72 ($72/12 = 6$), managers translate this into a tangible reality: *Any approved project must double its money every 6 years.*
- **Evaluating the Danger of Debt:** It quickly highlights the risk of high-interest financing. A credit card used to buy equipment at 18% annual interest will double the debt owed in just 4 years ($72/18 = 4$).

1.9 Chapter Summary

This introductory chapter established the foundational principles and terminology of engineering economy. The core objective of this discipline is to evaluate capital investment alternatives using structured, mathematically sound methodologies.

Key takeaways from this chapter include:

- **The Engineer's Role:** Engineers must look beyond physical design to evaluate the economic viability of projects, systematically comparing alternatives to maximize future value.
- **The Time Value of Money:** The most critical concept in financial decision-making; a dollar today is worth more than a dollar tomorrow due to its earning potential (interest).
- **Interest and Equivalence:** *Compound interest* assumes that accumulated interest earns further interest. Because of this compounding effect, different sums of

money at different points in time can be considered economically equivalent.

- **Cash Flow Diagrams:** A standardized graphical tool used to map cash inflows (upward arrows) and outflows (downward arrows) across a timeline, ensuring no costs or revenues are overlooked during analysis.
- **Minimum Attractive Rate of Return (MARR):** The “hurdle rate” or baseline profit margin a project must exceed to be considered viable. It is driven by the firm’s cost of capital, project risk, and opportunity costs.
- **The Rule of 72:** A practical mental math shortcut used to quickly estimate the time required for an investment to double at a given compound interest rate ($\text{Years} \approx 72/i\%$).